

Just as we can add on a graphics card using the PCI-e port of our computer, we can add a 'shield' using the port pins on the Arduino™ board. While graphics cards enhance the gaming experience or video editing capabilities of our computer, Arduino™ shields add extra functionality to the Arduino™ board.

Arduino™ acts as a platform for the development of many kinds of applications. The creation of such applications is supported by the addition of extra hardware such as sensors and motors connected to the pins on the board.

Why do we need Shields?

Many a time hardware can't be directly connected to the microcontroller. There are limitations to the electric power, voltage etc. that a microcontroller can supply to the connected hardware. For example, the most common component to be connected, a toy DC (Direct Current) motor, uses anywhere between 20 to 1000 times the electric power that any microcontroller can provide. Connecting such components directly can damage the microcontroller. This is where shields come into picture.



Why Shield?

What's a Shield?

A 'shield' is basically a printed circuit board (PCB) with some integrated circuits (ICs) and other hardware connected in a manner that allows it to be directly placed on top of an Arduino™ board. Moreover, a shield also simplifies the development process by providing libraries for easy programming.

What if there's no shield available for my application?

It may happen that your application's hardware is not supported by any shield. This is an excellent learning opportunity. In such cases to simplify the development process, you can use the Arduino™ Proto shield, which is basically a through-hole (perforated) PCB. With a little help from Google, you'll be able to find sensors and other hardware that can be connected on the Proto shield.